New trading relationships can change the dominant vehicle.

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

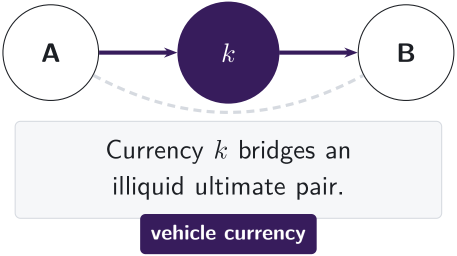
Jiahua Xu

A cross-border payment needs someone to make the FX market

THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

Project Nexus requires an FX provider. Project Rialto uses a vehicle when direct exchange is unavailable.



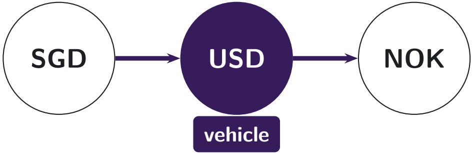
Note: Project Nexus states that every cross-border, cross-currency payment needs an actor willing and able to exchange the currencies. Project Rialto experimentally uses a vehicle currency when direct exchange is unavailable. The paper studies the intermediary asset observed inside DEX pool routes.

Pool routes reveal the vehicle currency

SINGAPORE PAYMENT ANALOGY

A Singapore-dollar payment to Norway may be converted through US dollars.

Conventional turnover data show two legs, not the customer's connected route.



Pool logs reveal the path.

DeFi is on-chain finance executed by smart contracts; decentralised exchanges (DEXs) trade tokens through liquidity pools. **Stablecoins** are tokens designed to track a reference currency, usually the US dollar.

Note: The currency labels illustrate a possible indirect conversion, not an observed customer route. Conventional FX records report turnover in each currency pair but generally do not link the two legs to one instruction. The DEX setting preserves the transaction-level pool path, so the intermediary asset is observed directly.

The route panel links pool-level swaps

472 million
pool-level swaps

2,332
calendar dates

8
DEX deployments

February 2020–June 2026

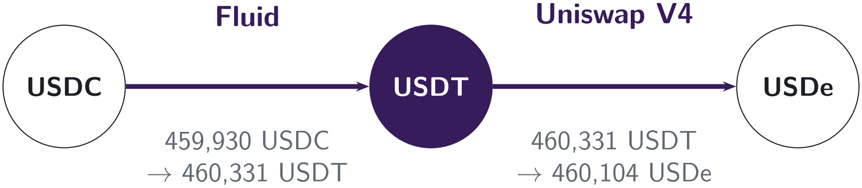
Uniswap v2, v3, and v4; SushiSwap v2 and v3; Curve; Balancer; and Fluid.

Note: The listed deployments form the principal Ethereum route panel. Uniswap v1 enters the separate protocol-architecture analysis because its retained records do not recover ultimate-pair identities.

Inside the pools, USDT links USDC to USDe

ORDERED TRANSFERS AND ROUTE RECONSTRUCTION

User instruction: PYUSD $\rightarrow$ USDe $\Longrightarrow$ Pool route: USDC $\rightarrow$ USDT $\rightarrow$ USDe



vehicle

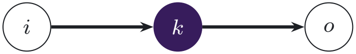
The ultimate pair is USDC $\rightarrow$ USDe. Its atomic pairs are USDC–USDT and USDT–USDe; USDT is the vehicle that links them.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026. The pool component begins at USDC. It does not reveal whether PYUSD became USDC in an unobserved pool or through the executor's inventory.

The full route and the one-vehicle choice answer different questions

MEASUREMENT

- Every complete route contributes each currency used in an intermediary position.
- Exact two-atomic-trade routes isolate one vehicle choice.

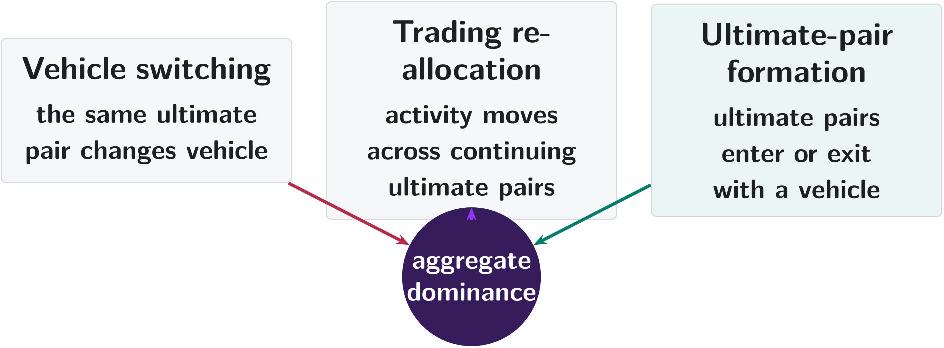


All route lengths
Presence on a route and share of all intermediary positions

One vehicle per route
Exact two-atomic-trade routes support decomposition and choice tests

Note: Route-participation shares can sum above 100% because a longer route may use several intermediary currencies. Intermediary-position shares divide all realised intermediary uses and sum to 100%. Exact two-atomic-trade routes map one route to one vehicle.

Aggregate dominance can change in three ways



Note: The decomposition is exact. The next step asks why an ultimate pair keeps its first vehicle and when a competing two-atomic-pair bridge becomes deep enough to attract routes.

Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE

217,003 token-to-token trades pass through ETH

Every pool is an ETH–token pool, so the architecture imposes the vehicle.

$\Longrightarrow$

V2: ULTIMATE-PAIR FORMATION

95.5%

of single-leg V2 trades use a WETH pool in 2026

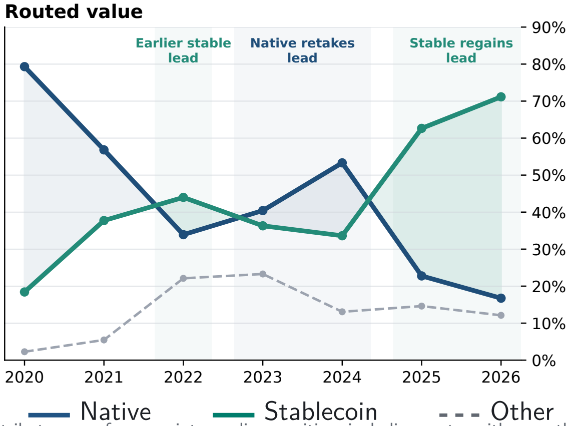
97.9%

of ultimate pairs first traded in 2026 include WETH

Note: V1 links token-to-ETH and ETH-to-token swaps within one transaction. V2 reports atomic trades and newly traded ultimate pairs on that venue. Early V2 inherited liquidity, users, and routing conventions from V1, so part of the initial persistence is mechanical; later ultimate-pair formation shows how long that structure lasts.

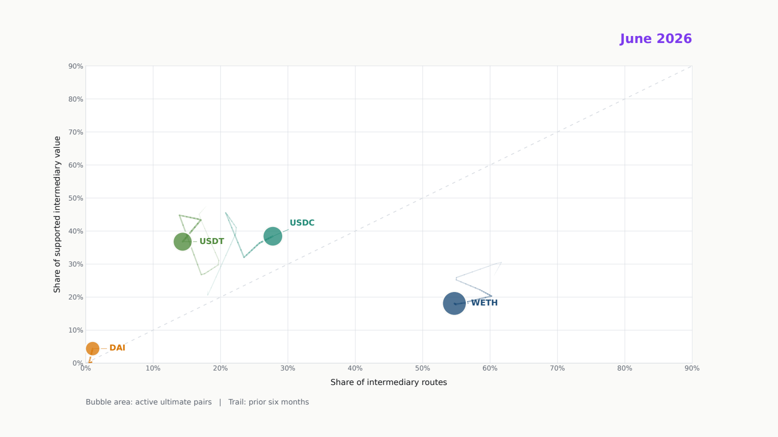
Stablecoins regain the routed-value lead by 2026 H1

ALL COMPLETE ROUTE LENGTHS



Note: Each complete route contributes once for every intermediary position, including routes with more than two atomic trades. Historical points through 2025 use full calendar years; 2026 covers January through June. Routed value requires source, intermediary, and destination values to agree within 20%.

Vehicle leadership turns over through time



▸ Play 18-second film

Note: The film follows WETH, USDC, USDT, and DAI in route participation, supported value, and ultimate-pair breadth. Each frame retains six months of history, so no single frame contains the full transition. The poster is the PDF fallback.

Ultimate-pair formation supplies most of the rotation

EXACT DECOMPOSITION IN PERCENTAGE POINTS (PP): 2024 H1 TO 2026 H1

Same continuing ultimate pair	Trading reallocation	Ultimate pairs in one period
-0.1 **pp**	$+8.6$ **pp**	$+17.8$ **pp**

Stablecoin share

2024 H1: **16.9%** → 2026 H1: **42.5%**

The aggregate rises $+25.7$ pp; net switching inside continuing ultimate pairs is -0.1 pp.

Note: Exact two-atomic-trade native-or-stable vehicle routes on matched January–June dates. A further -0.5 pp comes from the changing weight of continuing versus period-specific ultimate pairs. Components use unrounded values and sum to the aggregate change.

The vehicle chosen at ultimate-pair entry predicts later use

ULTIMATE PAIRS WITH COMPLETE FOLLOW-UP SUPPORT

INITIAL STABLECOIN SHARE
+1 pp
at ultimate-pair entry

$\Longrightarrow$

NEXT 30 DAYS
$+0.86$ **pp**
stablecoin share

$\Longrightarrow$

NEXT 120 DAYS
$+0.74$ **pp**
stablecoin share

The first vehicle becomes a durable focal point for the new trading relationship.

Note: 599,870 non-WETH ultimate-pair–route-scope observations. Regressions control for entry size, cohort year, endpoint class, direct-route share, and route complexity. Standard errors are clustered by entry date. Initial vehicle use is observed, not randomly assigned.

A shallow stablecoin bridge attracts little route flow

FIRST 30 DAYS AFTER PERSISTENT STABLECOIN SUPPORT

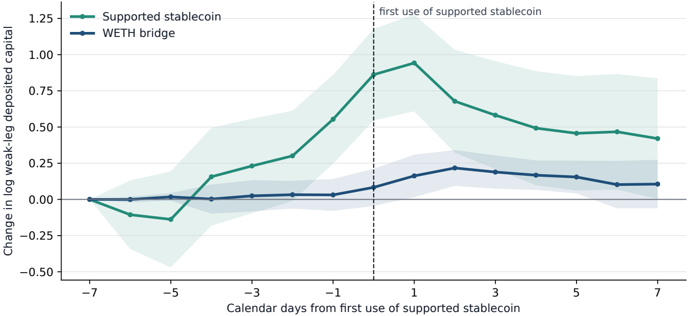
Below $0.1\times$ WETH depth	At least WETH depth	At least $2\times$ WETH depth
2.4%	**53.0%**	**69.9%**

First-month use rises from 42.6% below $0.1\times$ WETH depth to 84.1% above it.

Note: Depth is prior-calendar deposited capital on the weaker atomic pair. The route-share comparison uses 11,327 ultimate-pair days across 855 bridge events. Support, depth, and first use refer to the same stablecoin set. Capital and route demand are jointly determined.

Bridge capital builds before first use

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



$+0.86$

stablecoin bridge, day -7 to 0

$+0.78$

relative to the matched WETH bridge

Capital on the weaker atomic pair rises before the route appears.

Note: Equal-event path for 246 ultimate-pair events whose supported stablecoin is first used 7 to 119 days after persistent support. Capital is lagged one calendar day; shading gives 95% confidence intervals clustered by ultimate pair and adoption date. The sample conditions on eventual use.

Price competition usually changes the venue

SAME INPUT AND PRE-TRANSACTION STATE

USED VENUES

6.6%

better same-vehicle quote

ALL EXACT VENUES

44.4%

better same-vehicle quote

ALL VEHICLES + DIRECT

46.4%

better route

2.0 pp

added by opening vehicle identity and the direct path

-1.2 pp

change in stablecoin vehicle share

Note: 777,651 common-support routes on 73 monthly dates. Exact Uniswap v2, SushiSwap v2, and Uniswap v3 states; at least $100 input; 5% own-price-impact limit. The median gain among improved routes is 21.9 bp. Improvements are gross of gas.

What changes a dominant vehicle?

1. New relationships

Ultimate-pair entry and trading reallocation carry the aggregate rotation.

2. A first vehicle

The vehicle used at entry predicts the same ultimate pair's later routes.

3. Two deep legs

A challenger attracts flow when both atomic pairs become competitive.

Dominance changes when the trading network grows around a liquid vehicle.

Price competition is active across venues; vehicle identity moves much less.

Appendix map

Definitions and data

A1 Route units and vehicle status
A2 Asset types and backing regimes
A3 Sample construction and exclusions

Reconstruction and validation

A4 Exact transaction state
A5 Protocol-specific state
A6 Forced routes in V1

Robustness

A7 Timing and venue scope
A8 Count and value weighting
A9 Observed route endpoints

Market structure

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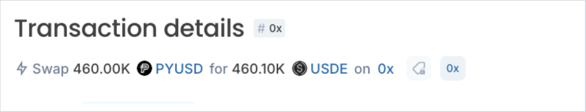
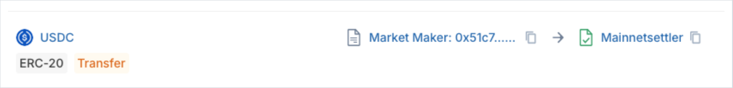
A18 Local and network depth
A19 Pool activation and growth
A20 V4 participation

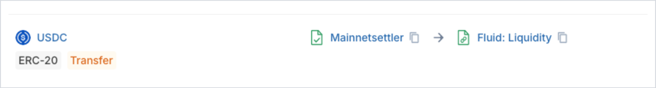
Pool data may start after the user instruction

USER INSTRUCTION RECORDED BY THE EXPLORER



THE DECISIVE SETTLEMENT TRANSFERS

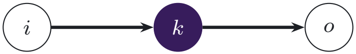




The maker supplies the USDC used by the Fluid leg. The instruction is PYUSD → USDe.

Note: Transaction `0xbda1d07f...9bf67ad9`, 10 January 2026. The data do not reveal whether PYUSD–USDC is another vehicle leg or executor inventory. Disconnected groups comprise 6.2% of reconstructed components in 2026. Treating each internally valid component as a separate route raises stablecoin count dominance from 42.3% to 43.7%.

A1. One reconstructed route is one unit

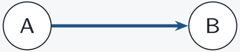


one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

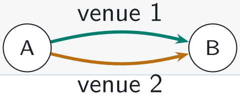
How we reconstruct a route from one transaction

Direct swap
one transaction

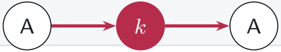


Sequential vehicle route
one transaction

Parallel split
one transaction

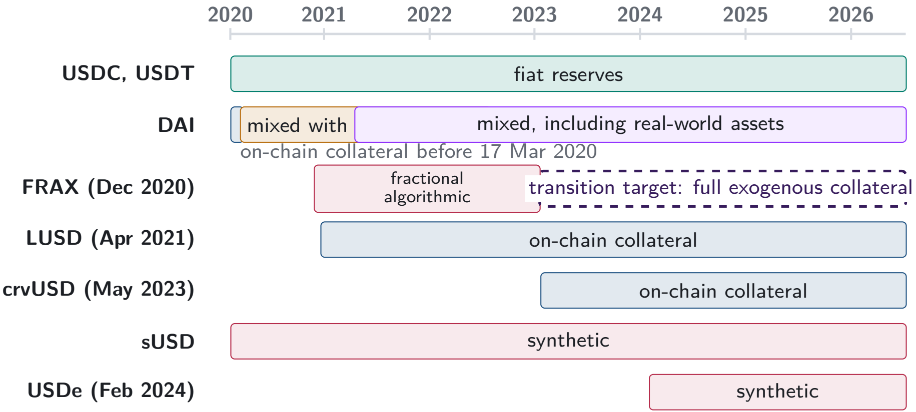


Round trip
one transaction



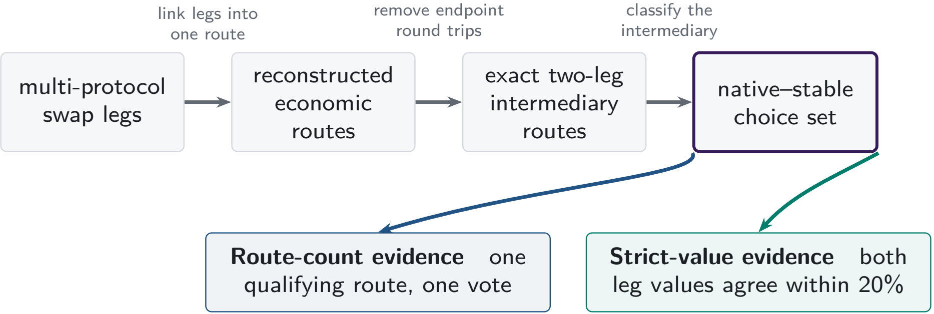
Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths make that choice visible inside aggregate turnover.

A2. Stablecoin backing changes over time



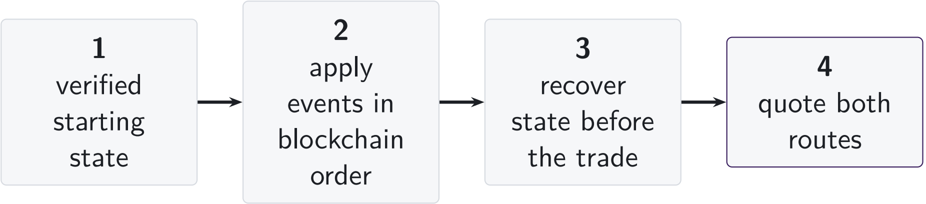
Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

A3. One route universe supports two measurements



Atomic token pairs inside multi-asset pools can supply route legs. V1/V2 architecture evidence is separate; dollar support changes value weighting.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hook-dependent pricing
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

Executable-quote validation

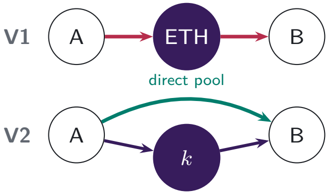
■ recovered pricing input
□ hook-dependent pricing unavailable

Protocol mechanisms studied

V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.

A5.1. Pool formation determines available paths

V1 PROTOCOL RULE $\longrightarrow$ V2 MARKET CHOICE



V1 admits only ETH–token pools. V2 permits any ERC-20 pool, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth comes from active liquidity

V3 CONCENTRATED-LIQUIDITY POSITIONS

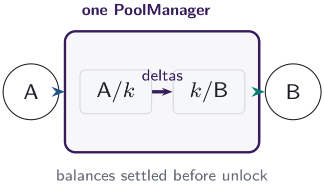
LPs choose a price range and fee tier for each atomic pool. Positions covering the current price determine active liquidity in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



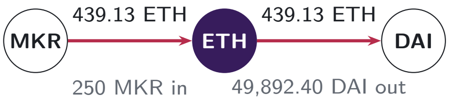
One PoolManager holds all pools. During a lock, the two route legs update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes the settlement boundary.

A6. V1 forced routes have an on-chain signature

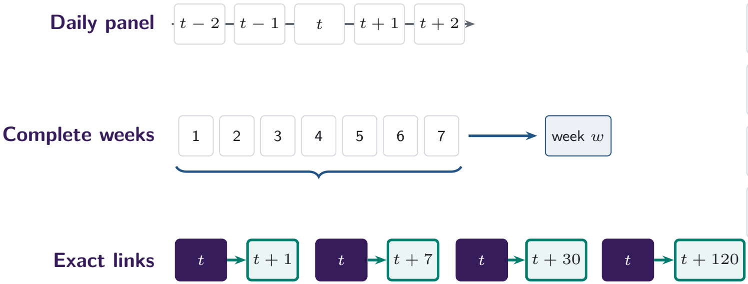
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

Note: Both legs of transaction `0x4dca160d...96a0ca16` (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

A7. Daily and weekly frequencies answer different questions

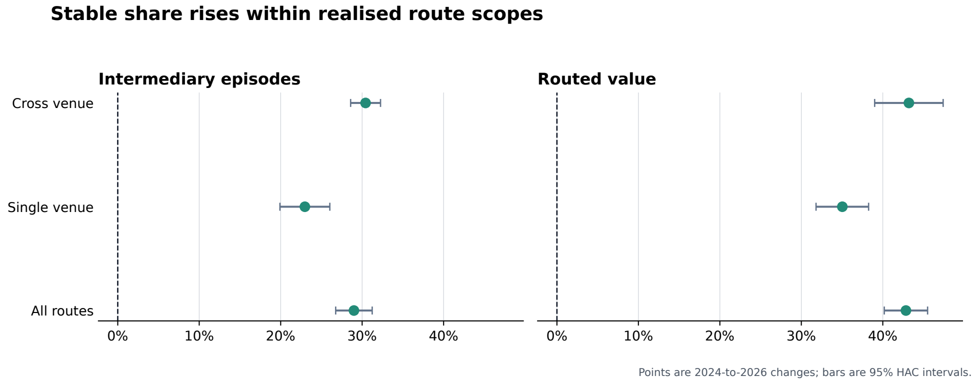


Exact horizons use the required target calendar date.

Model role

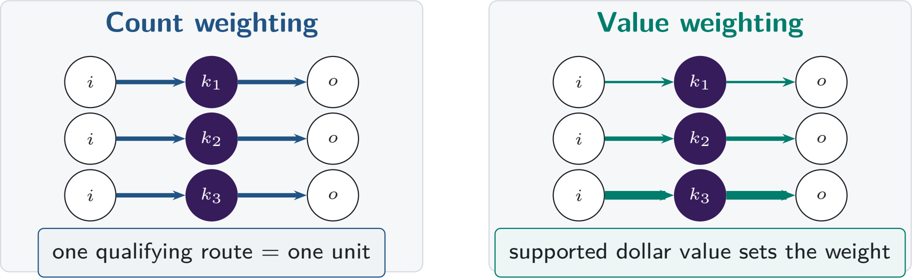
- **Daily panel** date fixed effects
- **Calendar check** week and weekday fixed effects
- **Complete weeks** week fixed effects
- **Exact links** inference by origin date

A7b. Stablecoin use rises within every venue scope



Note: Changes compare matched January-through-June dates in 2024 and 2026. Bars show calendar-HAC confidence intervals; venue scope records realised routes.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

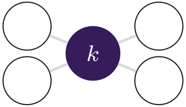
D_k = use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

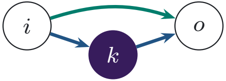
Excess vehicle use

$X_k = D_k \div E_k$
E_k: endpoint share of k

$X_k > 1$: use exceeds the endpoint benchmark



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I^N_{k,t} = \frac{n^I_{k,t}}{\sum_j n^I_{j,t}}$$

How often asset k sits inside a route.

Endpoint share

$$E^N_{k,t} = \frac{n^E_{k,t}}{\sum_j n^E_{j,t}}$$

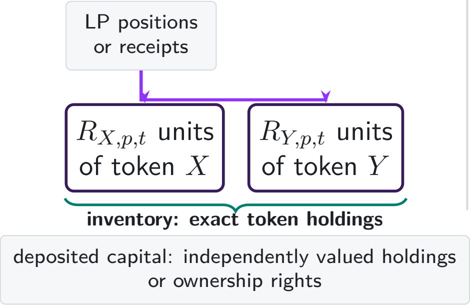
How often asset k appears at either end of the observed pool route.

Observed pool-route endpoints supply the benchmark.

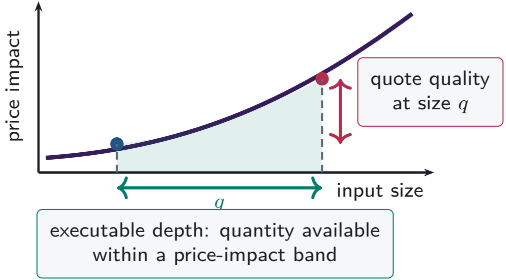
Excess use compares intermediation with observed route-endpoint use on the same support.

A10. Capital, inventory, and depth differ

Pool state at time t



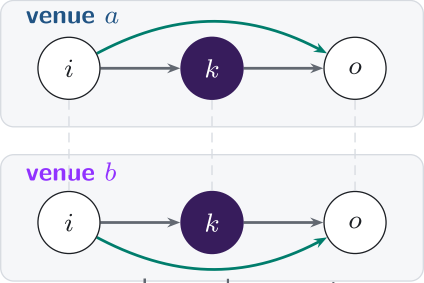
Execution from that state



Depth and quotes vary with size and state.

A11. Additional venues expand the feasible route set

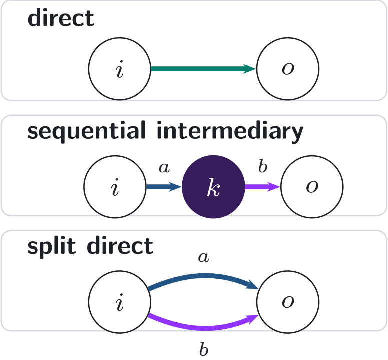
Feasible venue layers



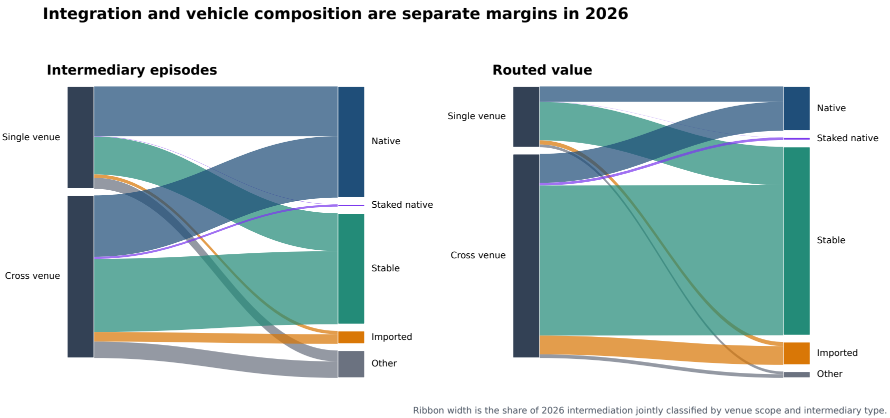
venue layers share assets;
available links define reach

realised
execution

Realised route topology



A11b. Venue scope and vehicle type differ in 2026



Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage

A14. WETH stays the graph hub as realised use shifts

ALL ROUTE LENGTHS; MONTHLY ATOMIC-PAIR GRAPHS

WETH BETWENNESS

#1

in all 7 annual graphs

0.927

in 2026 H1

WETH REALISED SHARE

76.2%

2024

42.3%

2026 H1

USDC + USDT REALISED SHARE

14.9%

2024

37.4%

2026 H1

Binary network position changes much less than actual intermediary use.

Note: Betweenness is approximate and unweighted in annual atomic-pair graphs built from trades on the fifteenth day of each month, using 256 source nodes and a fixed seed. Realised shares use all intermediary positions on the same dates. A graph edge records an observed atomic pair, not its depth or execution cost.

A15. Endpoint demand and intermediary use move together

	Intermediary share $(I_{a,t})$ [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	$+34.56$	$+34.55$	$+0.05$	—
	(23.04)	(23.04)	(0.56)	—
Stablecoin	$+2.42^{*}$	$+2.42^{*}$	$+0.85^{*}$	—
	(1.32)	(1.32)	(0.50)	—
Endpoint-demand share $(D_{a,t})$	—	—	$+1.59^{***}$	$+0.98^{***}$
	—	—	(0.03)	(0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

After date and currency effects, endpoint demand passes through almost one for one into intermediary use.

Note: 1,828,862 currency-days over 2,259 dates. Parentheses report currency/date-clustered standard errors. The outcome and demand regressor are in pp. Currency effects absorb time-invariant asset-class indicators. Asterisks $*$, $**$, and $***$ denote significance at 10%, 5%, and 1%, respectively.

A16. Ultimate-pair composition contains a mechanical component

WETH

WETH is an endpoint

↓

Eligible intermediaries: stablecoins

$+21.1$ **pp**

All matched ultimate pairs

WETH-linked route-count mass: 26.6% $\to$ 46.8%

$+3.7$ **pp**

Ultimate pairs without WETH endpoints

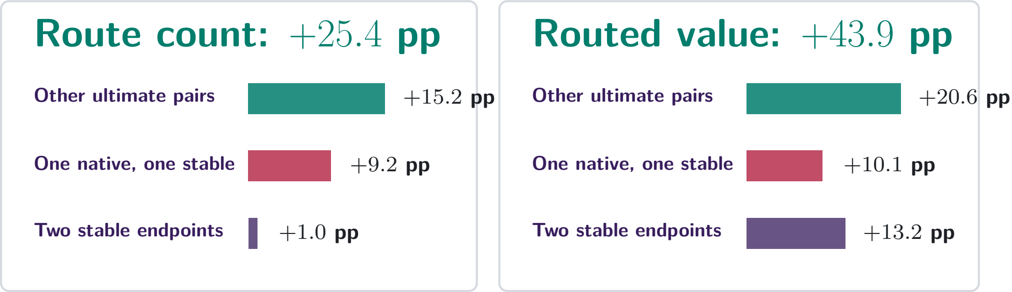
79,347 matched groups

Changes within fixed ultimate pairs are $+0.2$ pp overall and $+0.3$ pp after removing WETH endpoints.

Note: WETH cannot be both an ultimate-pair endpoint and the intermediary. Exact two-atomic-trade native-WETH-versus-stablecoin routes on common month-days; fixed-pair estimates also absorb month-day and route scope.

A17. Endpoint direction separates count and value channels

CONTRIBUTION TO THE STABLECOIN-SHARE INCREASE



USDT supplies $+13.7$ pp of the $+13.2$ pp two-stable-endpoint value channel.

Note: Exact additive decomposition on 181 matched dates in 2024 and 2026. Endpoint direction describes the ultimate pair; the vehicle remains the intermediary token.

A18. Local depth remains informative alongside network reach

84.1%
of routes use the deepest supported bridge

$+6.79$ **pp**
route share per log point of weak-leg capital

$+9.10$ **pp**
route share per log point of same-day network reach

Local two-atomic-pair depth and broader network position both matter.

Note: Five vehicle currencies compared within the same ultimate-pair–date–route-scope set. Depth is prior-calendar deposited capital on the weaker atomic pair. The regression absorbs set and vehicle effects and clusters by ultimate pair and date.

A19. Most first-use capital sits in pools already active

SUPPORTED STABLECOIN, DAY -7 TO FIRST USE

92.5%
of first-use capital lies in continuing pools

$+0.16$
continuing-pool growth relative to WETH

$+6.9$ **pp**
greater probability of a newly active pool

Note: 246 first-use events. A newly active pool has positive prior-calendar capital on day zero and none on day -7; continuing pools are positive on both dates. Difference inference clusters by ultimate pair and adoption date.

A20. V4 participation broadens after internal routing

10 PP MORE INTERNAL SAME-ASSET ROUTING

DAYS 1 TO 30
$+0.086^{***}$
actions by previously active origins

DAYS 31 TO 120
$+0.153^{***}$
origins first active after measurement

PERSISTENT VOLATILITY
$+0.318^{***}$
additional later first-active-origin association

Note: V4 vehicle-days with 180 prior days used to classify origins. Specifications absorb vehicle and date effects, include origin-day controls, and cluster by date. Transaction origins proxy for account participation. The estimates are predictive.